Lab - Preparing Data

Objectives

In this lab, you will learn the basics of cleaning data in Microsoft Excel. Data needs to be cleaned before it can be analyzed. Cleaning the data makes it easier to read and interpret, ensures consistency and accurate results, and enables a better decision-making process.

Part 1: Explore a Sample Set of Data

Part 2: Data Cleaning

Background / Scenario

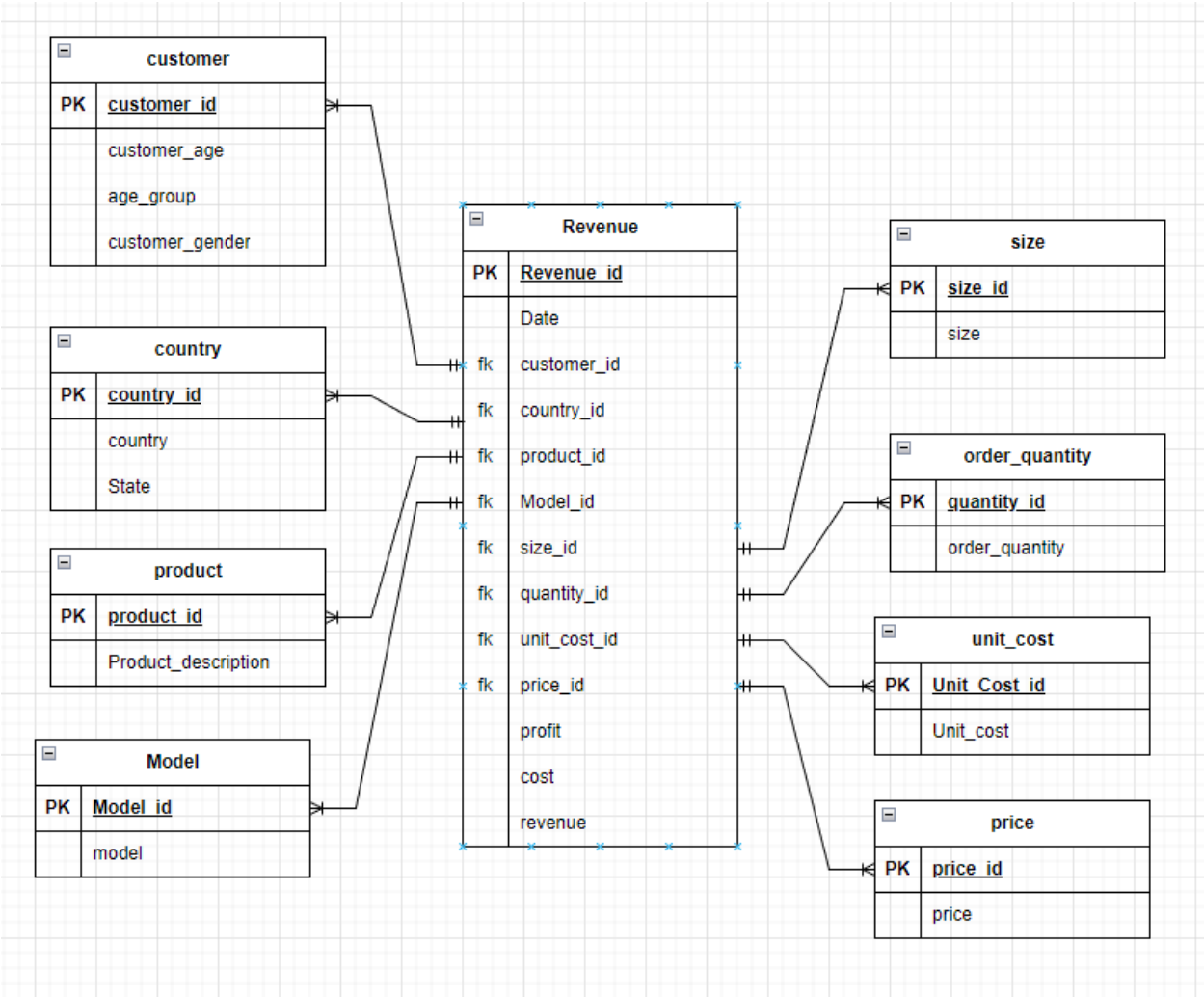
The rapidly increasing volume and complexity of data are due to growing mobile data traffic, cloud computing traffic, and adoption of technologies including the Internet of Things and Artificial Intelligence. Data in its raw form is useless unless it can be cleaned and analyzed to provide information that leads to knowledge and wisdom for actionable intelligence. Businesses are quickly discovering that they do not actually want data itself—they want the information and knowledge that comes from data, which they can use to make better decisions.

Required Resources

- Mobile device or PC/laptop with an internet connection and Microsoft Excel or similar spreadsheet program

Note: The precise steps to format and manipulate data in Excel can vary between platforms and versions. The instructions in this lab are based on the free version of Excel available from Office.com and may have to be modified to match the user's platform, software, or version to achieve the results shown in this lab.

Sales_Orde	Date	Day	Month	Year	Custome	Age_Group	Customer_(	Country	State	Produc	Sub_Categor	Model	Color	Size	Order_Quantit	Unit_Co	Unit_Pri	Profit	Cost	Revenue
000261695	01/12/2021	1	December	2021	39	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Black	46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261696	01/12/2021	1	December	2021	44	Adults (35-64)	Male	united kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261697	02/12/2021	2	December	2021	37	Adults (35-64)	Male	united states	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	2	$420.00	$769.00	$698.00	$840.00	$1,538.00
000261698	02/12/2021	2	December	2021	31	Young Adults (25-34)	Female	australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	1	$420.00	$769.00	$349.00	$420.00	$769.00
000261699	03/12/2021	3	December	2021	37	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Black	46	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261700	03/12/2021	3	December	2021	24	Youth (<25)	Female	united kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261701	03/12/2021	3	December	2021	37	Adults (35-64)	Male	united states	Washington	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261702	04/12/2021	4	December	2021	31	Young Adults (25-34)	Female	australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	4	$420.00	$769.00	$1,396.00	$1,680.00	$3,076.00
000261703	05/12/2021	5	December	2021	39	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Black	46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261704	05/12/2021	5	December	2021	42	Adults (35-64)	Male	germany	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200	Black	38	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261705	05/12/2021	5	December	2021	35	Adults (35-64)	Female	australia	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261706	05/12/2021	5	December	2021	37	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261707	06/12/2021	6	December	2021	23	Youth (<25)	Male	united kingdom	England	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	3	$420.00	$769.00	$1,047.00	$1,260.00	$2,307.00
000261708	06/12/2021	6	December	2021	27	Young Adults (25-34)	Male	canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261709	06/12/2021	6	December	2021	36	Youth (<25)	Male	australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261710	06/12/2021	6	December	2021	47	Adults (35-64)	Male	united kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261711	07/12/2021	7	December	2021	30	Young Adults (25-34)	Male	united states	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	4	$420.00	$769.00	$1,396.00	$1,680.00	$3,076.00
000261712	07/12/2021	7	December	2021	38	Adults (35-64)	Male	united states	California	Bikes	Mountain Bikes	Mountain-200	Silver	42	2	$1,266.00	$2,320.00	$2,108.00	$2,532.00	$4,640.00
000261713	08/12/2021	8	December	2021	19	Youth (<25)	Female	australia	New South Wales	Bikes	Mountain Bikes	Mountain-500	Silver	42	4	$308.00	$565.00	$1,028.00	$1,232.00	$2,260.00
000261714	08/12/2021	8	December	2021	30	Young Adults (25-34)	Female	canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Silver	38	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00	$9,280.00
000261715	08/12/2021	8	December	2021	39	Adults (35-64)	Female	united states	Oregon	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261716	08/12/2021	8	December	2021	35	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-500	Black	42	4	$295.00	$540.00	$245.00	$1,180.00	$2,160.00
000261717	09/12/2021	9	December	2021	33	Young Adults (25-34)	Female	australia	Victoria	Bikes	Mountain Bikes	Mountain-100	Black	38	2	$1,898.00	$3,375.00	$2,954.00	$3,796.00	$6,750.00
000261718	09/12/2021	9	December	2021	41	Adults (35-64)	Female	germany	Hamburg	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261719	10/12/2021	10	December	2021	34	Young Adults (25-34)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261720	10/12/2021	10	December	2021	40	Adults (35-64)	Male	australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261721	10/12/2021	10	December	2021	26	Young Adults (25-34)	Male	united kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261722	10/12/2021	10	December	2021	34	Young Adults (25-34)	Male	united states	California	Bikes	Mountain Bikes	Mountain-500	Black	40	1	$295.00	$540.00	$245.00	$295.00	$540.00
000261723	10/12/2021	10	December	2021	34	Young Adults (25-34)	Female	united states	Washington	Bikes	Mountain Bikes	Mountain-100	Silver	44	1	$1,912.00	$3,400.00	$1,488.00	$1,912.00	$3,400.00
000261724	10/12/2021	10	December	2021	38	Adults (35-64)	Male	australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261725	11/12/2021	11	December	2021	24	Youth (<25)	Female	france	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200	Black	38	3	$1,252.00	$2,295.00	$3,129.00	$3,756.00	$6,885.00
000261726	11/12/2021	11	December	2021	41	Adults (35-64)	Female	australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	2	$420.00	$769.00	$698.00	$840.00	$1,538.00
000261727	11/12/2021	11	December	2021	27	Young Adults (25-34)	Male	canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261728	11/12/2021	11	December	2021	37	Adults (35-64)	Male	united states	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	1	$420.00	$769.00	$349.00	$420.00	$769.00
000261729	11/12/2021	11	December	2021	38	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261730	12/12/2021	12	December	2021	36	Adults (35-64)	Female	australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Silver	42	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00	$9,280.00
000261731	12/12/2021	12	December	2021	37	Adults (35-64)	Male	united states	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	4	$420.00	$769.00	$1,396.00	$1,680.00	$3,076.00
000261732	12/12/2021	12	December	2021	34	Young Adults (25-34)	Male	australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00	$4,590.00
000261733	12/12/2021	12	December	2021	35	Adults (35-64)	Female	australia	Victoria	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261734	12/12/2021	12	December	2021	38	Adults (35-64)	Female	united states	Washington	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261735	13/12/2021	13	December	2021	32	Young Adults (25-34)	Female	australia	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	42	3	$1,266.00	$2,320.00	$3,162.00	$3,798.00	$6,960.00
000261736	13/12/2021	13	December	2021	40	Adults (35-64)	Female	united states	California	Bikes	Mountain Bikes	Mountain-500	Silver	40	1	$308.00	$565.00	$257.00	$308.00	$565.00
000261737	13/12/2021	13	December	2021	44	Adults (35-64)	Female	united kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00



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